

Mikhail Razakov

Compiler / Software Protection Engineer

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Compiler/program-analysis engineer with professional LLVM/static-analysis experience and a historical Unity licensing project. Strongest inspectable signals are LLVM legality/transformations, x86-64 codegen/register allocation, and Roslyn data flow; protection experience is deliberately bounded to server-side key activation, activation limits, and Unity-client integration.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's C#/.NET static-analysis platform.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
 - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.
- historical project**
• **date not claimed** **Unity licensing prototype**
- Server-side key issuance/activation, activation-count limits, and Unity-client integration; no anti-debugging or anti-tamper claim.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

SKILLS

Compiler / LLVM

C++23, LLVM 22/IR, interprocedural analysis, legality, transformations, LLVM Verifier

Program Analysis

Roslyn CFG, fixed-point data flow, state propagation/joins, conservative analysis

Low-Level / Codegen

Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

Software Protection / Licensing

server-side activation, activation-count limits, Unity integration

EDUCATION

HSE University — Software Engineering

RECOGNITION

HSE Faculty of Computer Science — open-source projects competition

Selected as one of the winners with UniversalToolchain / Wist2.

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture.

Earlier distinctions

Two-time overall winner of MEPhI Junior; prize-winner in HSE Vysshaya Proba competitive and industrial programming; First Degree Diploma and Grand Prize at the Baltic Science and Engineering Competition.

Moscow / remote